

LAPORAN PRAKTIKUM
SISTEM OPERASI
Perintah Dasar Linux

MODUL
13



Oleh:

(Alvin Bagus Firmansyah)

(2311104070)

PROGRAM STUDI S1 REKAYASA LUNAK
DIREKTORAT KAMPUS PURWOKERTO
UNIVERSITAS TELKOM

2025

BAB I

PENDAHULUAN

A. Dasar Teori

Linux adalah sistem operasi yang banyak digunakan pada komputer dan server. Linux dapat dioperasikan melalui terminal dengan menggunakan berbagai perintah untuk mengelola file, direktori, dan sistem. Beberapa perintah dasar yang sering digunakan antara lain ls, cd, cp, mv, dan rm. Linux juga memiliki fitur pipe dan redirection yang memudahkan pengolahan output suatu perintah. Selain itu, Linux mendukung proses pengembangan program dengan compiler GCC yang digunakan untuk mengubah source code bahasa C menjadi program yang dapat dijalankan. Dengan memahami perintah-perintah dasar Linux, pengguna dapat bekerja lebih efektif dalam mengelola sistem operasi.

B. Tujuan

Praktikum ini bertujuan untuk mengenal dan memahami penggunaan sistem operasi Linux melalui terminal. Selain itu, praktikum ini juga bertujuan agar mahasiswa dapat menggunakan perintah-perintah dasar Linux untuk mengelola file dan direktori, memahami penggunaan pipe dan redirection, serta melakukan kompilasi dan menjalankan program bahasa C menggunakan GCC.

BAB II

1. Terminal (4 poin)

Pada terminal Linux terdapat command prompt seperti ini:

```
ubuntu@localhost:~$
```

Catatan: ini hanyalah contoh, setiap komputer akan memiliki command prompt yang berbeda-beda.

Pada linux command prompt biasanya dalam bentuk

```
username@host : working_directory $
```

Maka, arti dari ubuntu@:~\$ adalah username “ubuntu” yang berada komputer bernama

“localhost”, ada pada direktori ~. Tanda \$ menunjukkan bahwa “ubuntu” adalah user biasa.

Jika root yang mengeksekusi maka \$ akan berubah menjadi #.

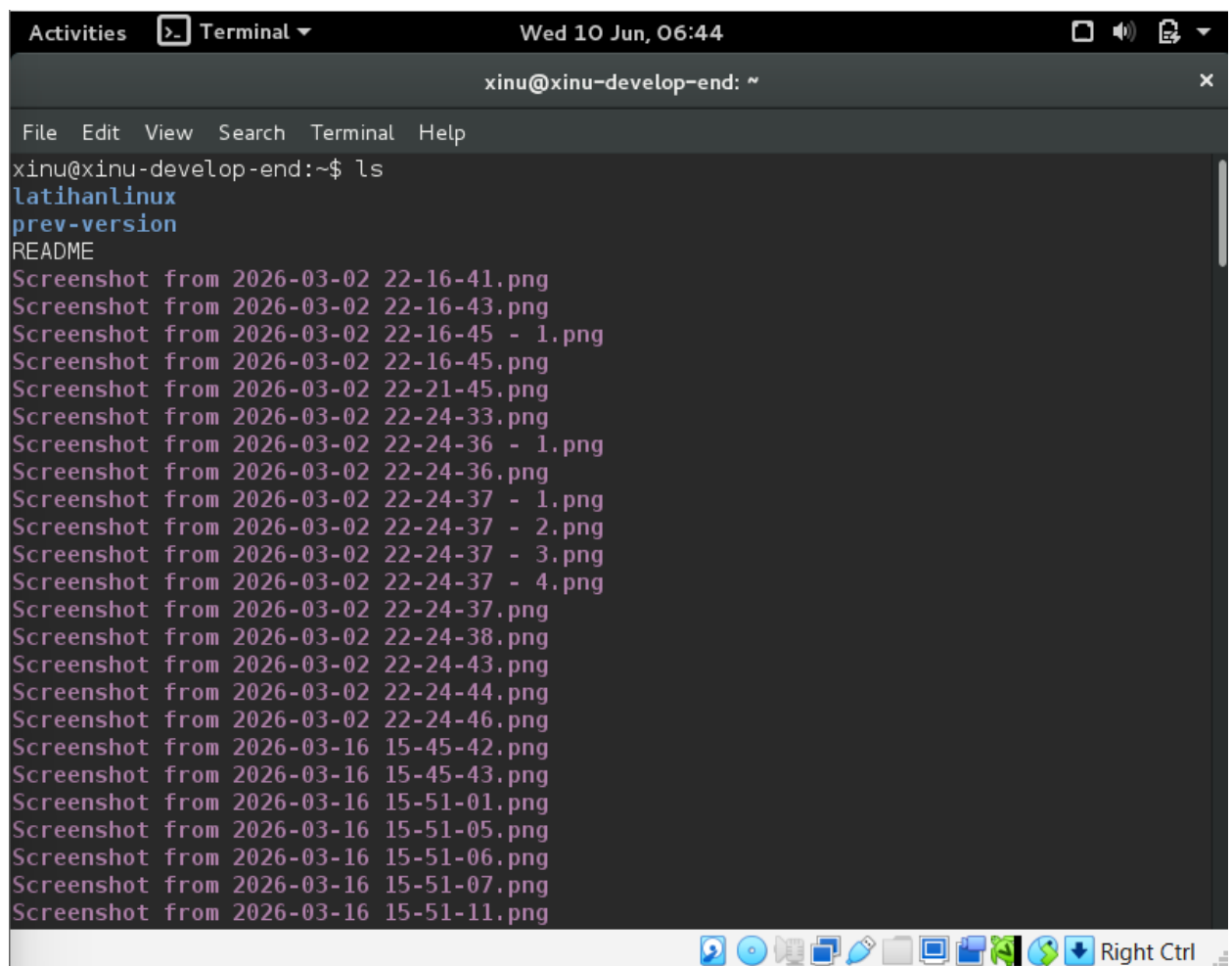
Pertanyaan:

- Jalankan dan screenshot terminal Anda!
- Jelaskan arti dari command prompt milik Anda!

Jawab:

Berdasarkan command prompt `xinu@xinu-develop-end:~$` maka artinya:

- `xinu` = username yang sedang login
- `xinu-develop-end` = nama komputer (hostname)
- `~` = direktori saat ini adalah home directory (`/home/xinu`)
- `$` = menunjukkan bahwa xinu adalah user biasa (bukan root). Jika root maka simbol \$ akan berubah menjadi #



```
Activities > Terminal Wed 10 Jun, 06:44
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ ls
latihanlinux
prev-version
README
Screenshot from 2026-03-02 22-16-41.png
Screenshot from 2026-03-02 22-16-43.png
Screenshot from 2026-03-02 22-16-45 - 1.png
Screenshot from 2026-03-02 22-16-45.png
Screenshot from 2026-03-02 22-21-45.png
Screenshot from 2026-03-02 22-24-33.png
Screenshot from 2026-03-02 22-24-36 - 1.png
Screenshot from 2026-03-02 22-24-36.png
Screenshot from 2026-03-02 22-24-37 - 1.png
Screenshot from 2026-03-02 22-24-37 - 2.png
Screenshot from 2026-03-02 22-24-37 - 3.png
Screenshot from 2026-03-02 22-24-37 - 4.png
Screenshot from 2026-03-02 22-24-37.png
Screenshot from 2026-03-02 22-24-38.png
Screenshot from 2026-03-02 22-24-43.png
Screenshot from 2026-03-02 22-24-44.png
Screenshot from 2026-03-02 22-24-46.png
Screenshot from 2026-03-16 15-45-42.png
Screenshot from 2026-03-16 15-45-43.png
Screenshot from 2026-03-16 15-51-01.png
Screenshot from 2026-03-16 15-51-05.png
Screenshot from 2026-03-16 15-51-06.png
Screenshot from 2026-03-16 15-51-07.png
Screenshot from 2026-03-16 15-51-11.png
```

2. Perintah pertama (7 poin)

Setiap perintah pada Linux mempunyai bentuk umum:

nama_perintah option command_line_argument

Apabila terdapat perintah: ls -al /

maka perintah tersebut berarti:

nama_perintah = ls

option (atau switch) = -al

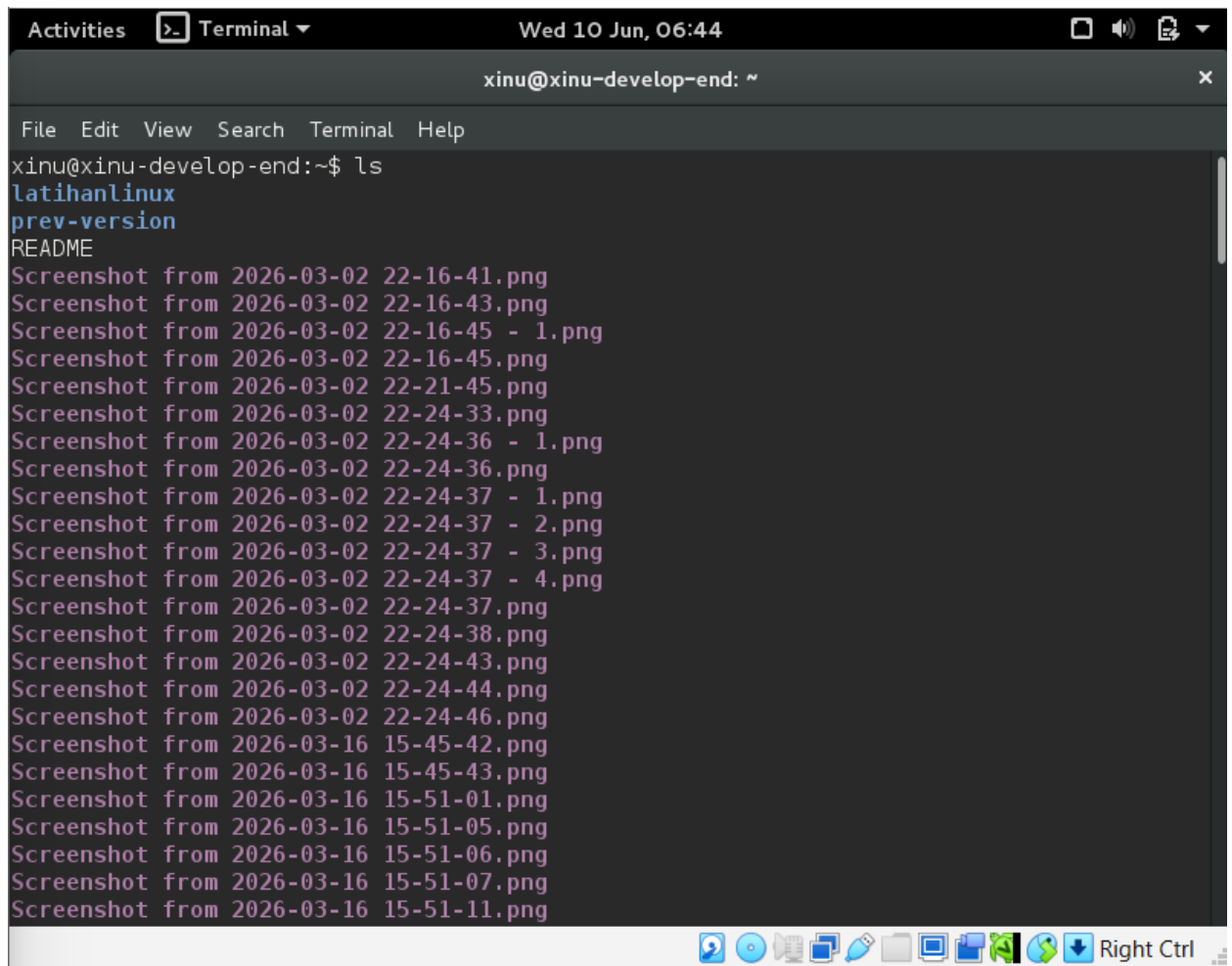
command_line_argument (atau parameter) = /

Catatan: beberapa perintah dapat berjalan tanpa adanya option maupun parameter, beberapa perintah hanya dapat berjalan jika diberikan parameter, perintah yang lain membutuhkan option dan parameter untuk dapat berjalan.

Perintah bersifat case sensitif: perintah ls berbeda dengan perintah LS

Pertanyaan:

- a. Jalankan dan screenshot perintah berikut ini: ls



```
xinu@xinu-develop-end:~$ ls
latihanlinux
prev-version
README
Screenshot from 2026-03-02 22-16-41.png
Screenshot from 2026-03-02 22-16-43.png
Screenshot from 2026-03-02 22-16-45 - 1.png
Screenshot from 2026-03-02 22-16-45.png
Screenshot from 2026-03-02 22-21-45.png
Screenshot from 2026-03-02 22-24-33.png
Screenshot from 2026-03-02 22-24-36 - 1.png
Screenshot from 2026-03-02 22-24-36.png
Screenshot from 2026-03-02 22-24-37 - 1.png
Screenshot from 2026-03-02 22-24-37 - 2.png
Screenshot from 2026-03-02 22-24-37 - 3.png
Screenshot from 2026-03-02 22-24-37 - 4.png
Screenshot from 2026-03-02 22-24-37.png
Screenshot from 2026-03-02 22-24-38.png
Screenshot from 2026-03-02 22-24-43.png
Screenshot from 2026-03-02 22-24-44.png
Screenshot from 2026-03-02 22-24-46.png
Screenshot from 2026-03-16 15-45-42.png
Screenshot from 2026-03-16 15-45-43.png
Screenshot from 2026-03-16 15-51-01.png
Screenshot from 2026-03-16 15-51-05.png
Screenshot from 2026-03-16 15-51-06.png
Screenshot from 2026-03-16 15-51-07.png
Screenshot from 2026-03-16 15-51-11.png
```

- b. Apakah option dan parameter dari perintah di atas?
- Option : tidak ada
 - Parameter : tidak ada
- c. Apa fungsi dari perintah tersebut?
- Perintah ls berfungsi untuk menampilkan daftar file dan folder yang ada di direktori saat ini
- d. Jalankan perintah berikut ini: ls -al /

```
xinu@xinu-develop-end:~$ ls -al /
total 88
drwxr-xr-x 21 root root 4096 Nov  4 2025 .
drwxr-xr-x 21 root root 4096 Nov  4 2025 ..
drwxr-xr-x  2 root root 4096 Jun 10 2015 bin
drwxr-xr-x  3 root root 4096 Jun 10 2015 boot
drwxr-xr-x 17 root root 3000 Jun 10 09:29 dev
drwxr-xr-x 137 root root 12288 Jun 10 09:29 etc
drwxr-xr-x  3 root root 4096 Jul  3 2015 home
lrwxrwxrwx  1 root root   29 Jun 10 2015 initrd.img -> /boot/initrd.img-3.16.0-4-586
drwxr-xr-x 18 root root 4096 Jun 10 2015 lib
drwx----- 2 root root 16384 Jun 10 2015 lost+found
drwxr-xr-x  3 root root 4096 Jun 10 2015 media
drwxr-xr-x  2 root root 4096 Jun 10 2015 mnt
drwxr-xr-x  2 root root 4096 Jun 10 2015 opt
dr-xr-xr-x 149 root root   0 Jun 10 09:29 proc
drwx----- 2 root root 4096 Jul  3 2015 root
drwxr-xr-x 22 root root  860 Jun 10 09:29 run
drwxr-xr-x  2 root root 4096 Jul  6 2015/sbin
drwxrwxr-x  3 root xinu 4096 Jul  2 2015 srv
dr-xr-xr-x 12 root root   0 Jun 10 09:29 sys
drwxrwxrwt 15 root root 4096 Jun 10 09:29 tmp
drwxr-xr-x 10 root root 4096 Jun 10 2015 usr
drwxr-xr-x 12 root root 4096 Jun 10 2015 var
lrwxrwxrwx  1 root root   25 Jun 10 2015 vmlinuz -> boot/vmlinuz-3.16.0-4-586
```

- e. Apakah option dan parameter dari perintah di atas?
- Option : **-al** (gabungan dari **-a** dan **-l**)
 - Parameter : **/** (direktori root)
- f. Apa fungsi dari perintah tersebut?
- Perintah **ls -al /** berfungsi untuk menampilkan seluruh isi direktori root / secara detail, termasuk file tersembunyi. Dimana:
- **-a** = menampilkan semua file termasuk file tersembunyi (diawali titik)
 - **-l** = menampilkan dalam format detail (ukuran, izin akses, tanggal, dll)
 - **/** = direktori yang ditampilkan adalah root
- g. Jelaskan mengapa perintah pada a dan e mempunyai hasil yang berbeda!
- Hasilnya berbeda karena perintah **ls** tanpa option dan parameter hanya menampilkan file dan folder di direktori saat ini secara sederhana. Sedangkan **ls -al /** menampilkan semua file termasuk file tersembunyi di direktori root (/) secara lengkap dan detail termasuk ukuran file, izin akses, pemilik, dan tanggal modifikasi.

3. Pohon file (6 poin)

Melihat struktur direktori dan file pada Linux sama seperti melihat file dan direktori (folder) yang berada pada GUI. Pada Linux terdapat direktori utama yaitu “/” (root). Di bawah direktori root terdapat berbagai direktori yang lain seperti: home, etc, usr, dll. Contoh: program firefox terdapat pada path **/usr/lib/firefox**, berarti untuk mengakses firefox kita berjalan dari root kemudian direktori “usr” dilanjutkan direktori “lib” baru dapat mengeksekusi firefox

Pertanyaan:

a. Jalankan dan screenshot perintah: pwd

```
xinu@xinu-develop-end:~$ pwd
/home/xinu
```

b. Apakah option dan parameter dari perintah tersebut?

- Option : tidak ada
- Parameter : tidak ada

c. Apa fungsi perintah tersebut?

Perintah pwd (*Print Working Directory*) berfungsi untuk menampilkan lokasi/path direktori tempat kita berada saat ini. Contoh hasilnya /home/xinu yang berarti kita sedang berada di direktori home milik user xinu.

4. Perpindahan (6 poin)

a. Jalankan dan screenshot perintah: cd /

```
xinu@xinu-develop-end:~$ cd /
```

b. Apakah option dan parameter dari perintah tersebut?

- Option : tidak ada
- Parameter : / (direktori root)

c. Apa yang dilakukan perintah tersebut?

Perintah cd / berfungsi untuk berpindah ke direktori root / yaitu direktori paling atas dalam struktur Linux.

5. Direktori khusus (6 poin)

~ disebut direktori home. Misalkan user bernama linus, mempunyai home directory /home/linus, dengan melakukan cd ~ maka sama dengan cd /home/linus.

.. merepresentasikan parent directory. Jika kita berada dalam folder /home dan kemudian mengeksekusi perintah cd .. berarti kita berpindah dari direktori /home ke parentnya yaitu direktori /

. merupakan current directory, pada gambar pohon di atas jika kita berada pada folder /home/john maka untuk mengakses school dapat digunakan perintah ./Documents/school

a. Lakukan dan screenshot perintah cd / kemudian lakukan perintah cd ~. Jelaskan hasil dari keduanya!

```
xinu@xinu-develop-end:/$ cd /
xinu@xinu-develop-end:/$ pwd
/
xinu@xinu-develop-end:/$ cd ~
xinu@xinu-develop-end:~$ pwd
/home/xinu
```

hasil `cd /` dan `cd ~`, lalu tulis:

- `cd /` → berpindah ke direktori root /
- `cd ~` → berpindah ke direktori home yaitu /home/xinu

b. Lakukan perintah `cd /proc/self`. Buatlah perintah menggunakan `cd ..` agar dapat berpindah ke direktori / (root). Berapa kali perintah `cd ..` harus dieksekusi?

Screenshot hasilnya!

```
xinu@xinu-develop-end:~$ cd /proc/self
xinu@xinu-develop-end:/proc/self$ pwd
/proc/self
xinu@xinu-develop-end:/proc/self$ cd ..
xinu@xinu-develop-end:/proc$ pwd
/proc
xinu@xinu-develop-end:/proc$ cd ..
xinu@xinu-develop-end:/$ pwd
/
```

Perintah `cd ..` harus dieksekusi 2 kali karena /proc/self memiliki 2 tingkat dari root (/proc → self), sehingga butuh 2 kali naik ke parent directory untuk sampai ke /

6. Copy, rename dan delete file (screenshot setiap tahapan!) (14 poin)

Hint: gunakan `cp`, `mv`, dan `rm`

a. Copylah file dari /proc/cpuinfo ke folder home Anda (/home/user/) menggunakan command pada terminal. Ganti user dengan username Anda

```
xinu@xinu-develop-end:/$ cp /proc/cpuinfo /home/xinu/
```

.

b. Tunjukkan menggunakan perintah bahwa file tersebut benar-benar telah dicopy ke folder home Anda.

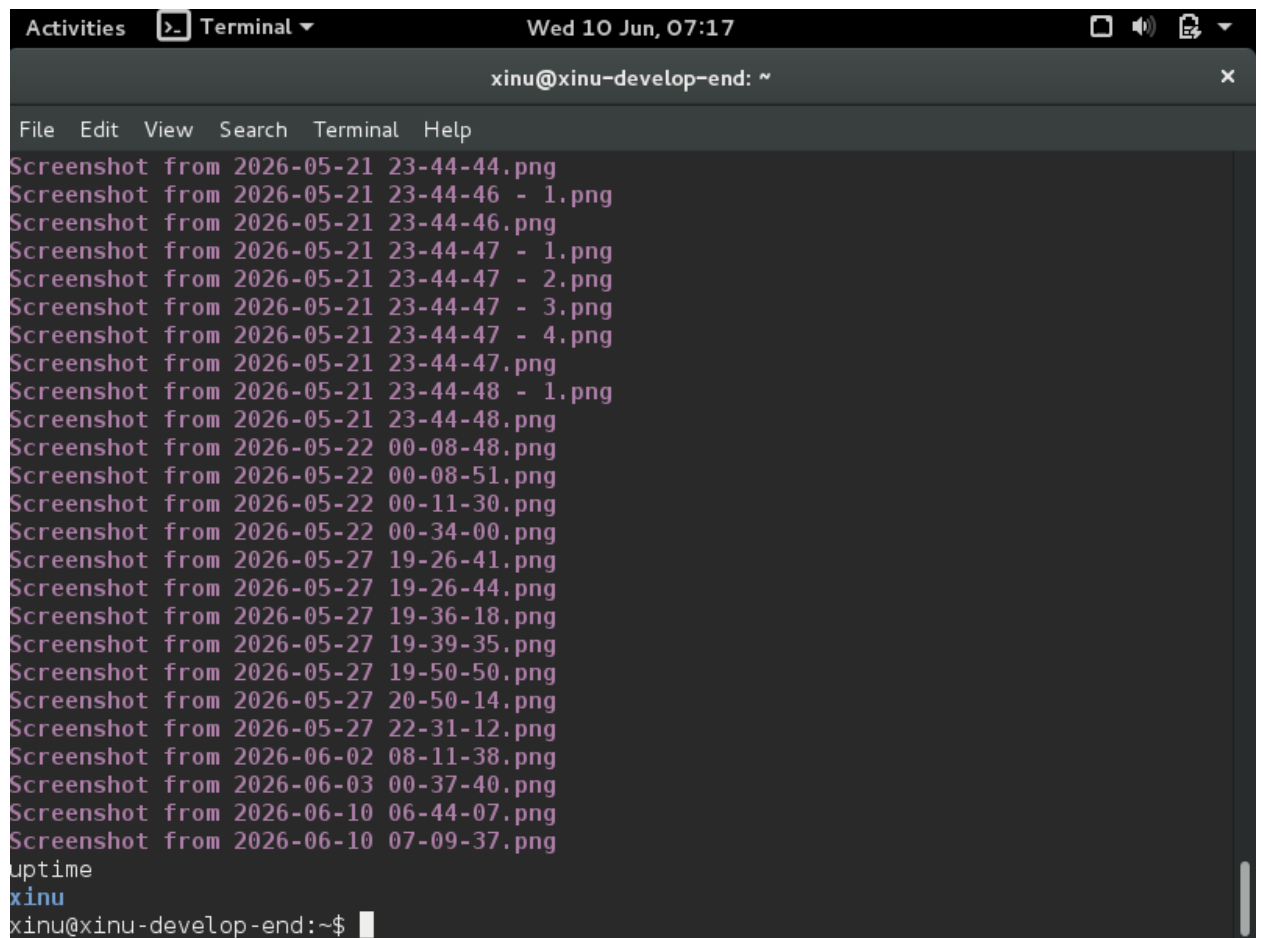
```
xinu@xinu-develop-end:/$ ls /home/xinu/
cpuinfo
latihanlinux
prev-version
README
Screenshot from 2026-03-02 22-16-41.png
Screenshot from 2026-03-02 22-16-43.png
Screenshot from 2026-03-02 22-16-45 - 1.png
Screenshot from 2026-03-02 22-16-45.png
Screenshot from 2026-03-02 22-21-45.png
```

c. Copy file dari /proc/uptime ke folder home Anda.

```
xinu@xinu-develop-end:~$ cp /proc/uptime /home/xinu/
```

d. Tunjukkan menggunakan perintah bahwa file tersebut benar-benar telah dicopy ke folder home Anda.


```
xinu@xinu-develop-end:~$ ls /home/xinu
cpuinfo
latihanlinux
prev-version
README
Screenshot from 2026-03-02 22-16-41.png
Screenshot from 2026-03-02 22-16-43.png
Screenshot from 2026-03-02 22-16-45 - 1.png
Screenshot from 2026-03-02 22-16-45.png
Screenshot from 2026-03-02 22-21-45.png
Screenshot from 2026-03-02 22-24-33.png
Screenshot from 2026-03-02 22-24-36 - 1.png
Screenshot from 2026-03-02 22-24-36.png
Screenshot from 2026-03-02 22-24-37 - 1.png
Screenshot from 2026-03-02 22-24-37 - 2.png
Screenshot from 2026-03-02 22-24-37 - 3.png
Screenshot from 2026-03-02 22-24-37 - 4.png
Screenshot from 2026-03-02 22-24-37.png
Screenshot from 2026-03-02 22-24-38.png
Screenshot from 2026-03-02 22-24-43.png
Screenshot from 2026-03-02 22-24-44.png
Screenshot from 2026-03-02 22-24-46.png
Screenshot from 2026-03-16 15-45-42.png
Screenshot from 2026-03-16 15-45-43.png
Screenshot from 2026-03-16 15-51-01.png
Screenshot from 2026-03-16 15-51-05.png
Screenshot from 2026-03-16 15-51-06.png
```



```
Activities Terminal Wed 10 Jun, 07:17 xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
Screenshot from 2026-05-21 23-44-44.png
Screenshot from 2026-05-21 23-44-46 - 1.png
Screenshot from 2026-05-21 23-44-46.png
Screenshot from 2026-05-21 23-44-47 - 1.png
Screenshot from 2026-05-21 23-44-47 - 2.png
Screenshot from 2026-05-21 23-44-47 - 3.png
Screenshot from 2026-05-21 23-44-47 - 4.png
Screenshot from 2026-05-21 23-44-47.png
Screenshot from 2026-05-21 23-44-48 - 1.png
Screenshot from 2026-05-21 23-44-48.png
Screenshot from 2026-05-22 00-08-48.png
Screenshot from 2026-05-22 00-08-51.png
Screenshot from 2026-05-22 00-11-30.png
Screenshot from 2026-05-22 00-34-00.png
Screenshot from 2026-05-27 19-26-41.png
Screenshot from 2026-05-27 19-26-44.png
Screenshot from 2026-05-27 19-36-18.png
Screenshot from 2026-05-27 19-39-35.png
Screenshot from 2026-05-27 19-50-50.png
Screenshot from 2026-05-27 20-50-14.png
Screenshot from 2026-05-27 22-31-12.png
Screenshot from 2026-06-02 08-11-38.png
Screenshot from 2026-06-03 00-37-40.png
Screenshot from 2026-06-10 06-44-07.png
Screenshot from 2026-06-10 07-09-37.png
uptime
xinu
xinu@xinu-develop-end:~$
```

e. Hapuslah file uptime di folder home Anda.

```
xinu@xinu-develop-end:~$ rm /home/xinu/uptime
rm: remove write-protected regular file '/home/xinu/uptime'? yes
```

f. Tunjukkan menggunakan perintah bahwa file tersebut benar-benar telah dihapus ke folder home Anda.

```
xinu@xinu-develop-end:~$ ls /home/xinu/
cpuinfo
latihanlinux
prev-version
README
Screenshot from 2026-03-02 22-16-41.png
Screenshot from 2026-03-02 22-16-43.png
Screenshot from 2026-03-02 22-16-45 - 1.png
Screenshot from 2026-03-02 22-16-45.png
Screenshot from 2026-03-02 22-21-45.png
Screenshot from 2026-03-02 22-24-33.png
Screenshot from 2026-03-02 22-24-36 - 1.png
Screenshot from 2026-03-02 22-24-36.png
Screenshot from 2026-03-02 22-24-37 - 1.png
Screenshot from 2026-03-02 22-24-37 - 2.png
Screenshot from 2026-03-02 22-24-37 - 3.png
Screenshot from 2026-03-02 22-24-37 - 4.png
Screenshot from 2026-03-02 22-24-37.png
Screenshot from 2026-03-02 22-24-38.png
Screenshot from 2026-03-02 22-24-43.png
Screenshot from 2026-03-02 22-24-44.png
Screenshot from 2026-03-02 22-24-46.png
Screenshot from 2026-03-16 15-45-42.png
```

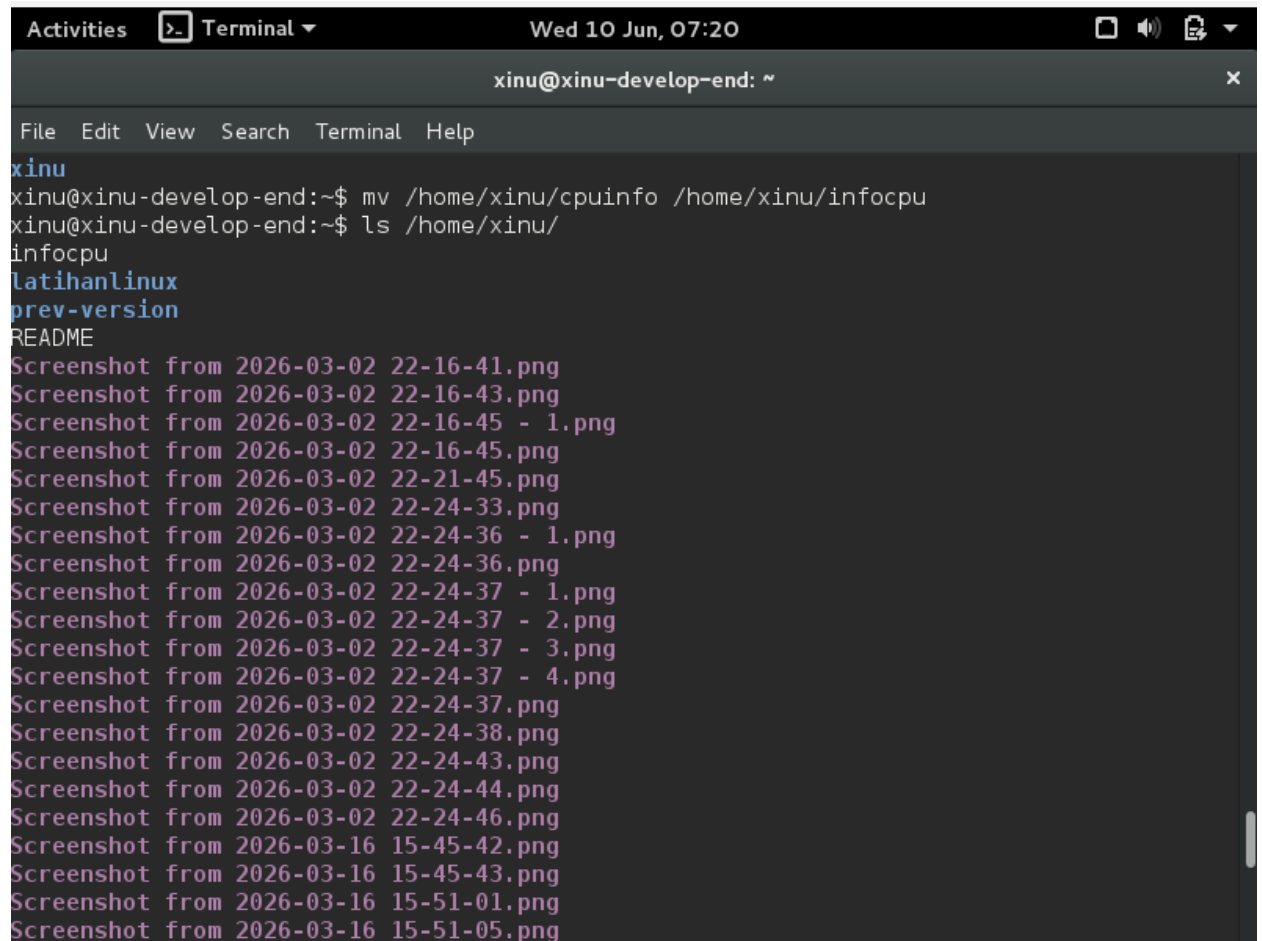
Activities  Terminal Wed 10 Jun, 07:19 

xinu@xinu-develop-end: ~

File Edit View Search Terminal Help

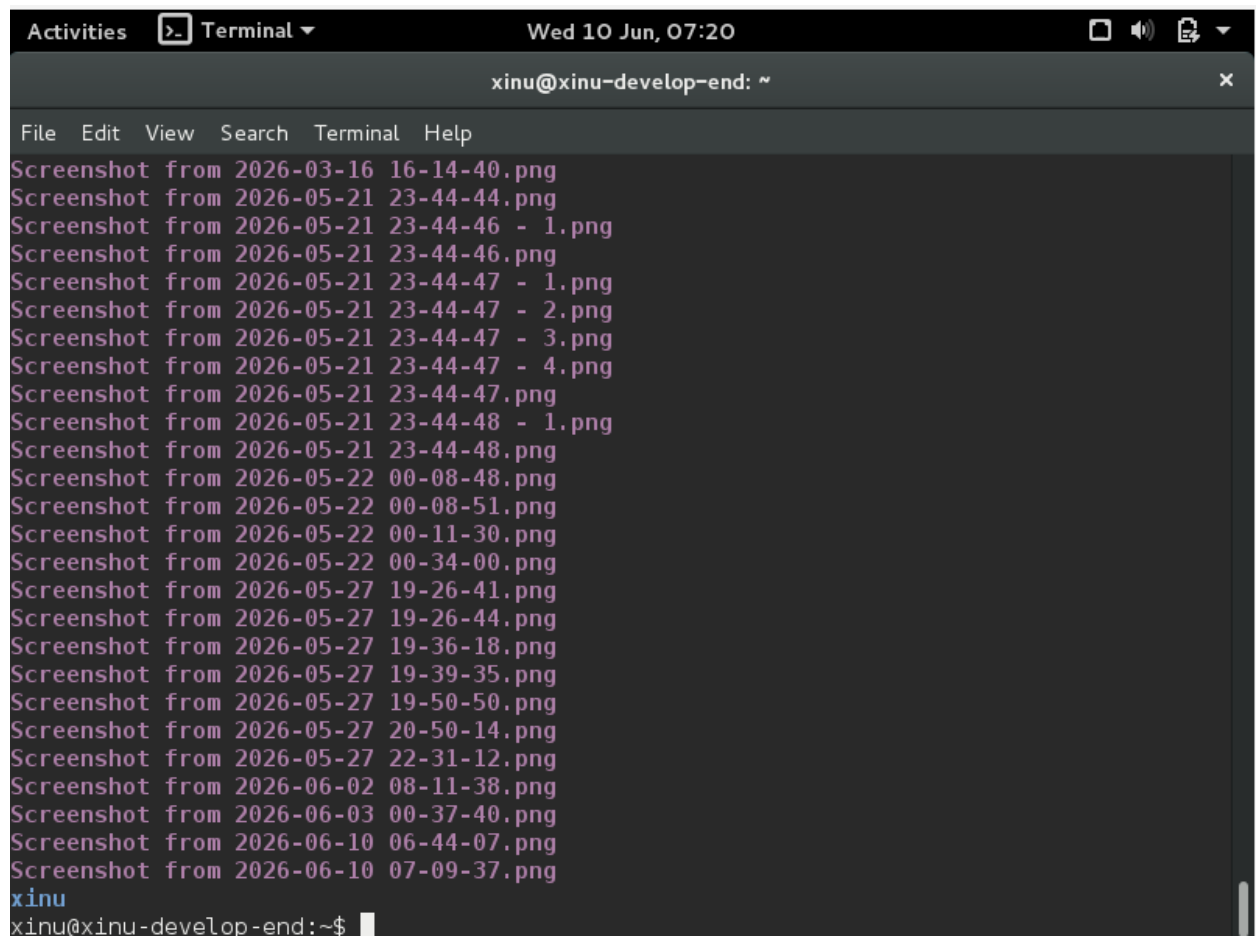
```
Screenshot from 2026-03-16 16-14-40.png
Screenshot from 2026-05-21 23-44-44.png
Screenshot from 2026-05-21 23-44-46 - 1.png
Screenshot from 2026-05-21 23-44-46.png
Screenshot from 2026-05-21 23-44-47 - 1.png
Screenshot from 2026-05-21 23-44-47 - 2.png
Screenshot from 2026-05-21 23-44-47 - 3.png
Screenshot from 2026-05-21 23-44-47 - 4.png
Screenshot from 2026-05-21 23-44-47.png
Screenshot from 2026-05-21 23-44-48 - 1.png
Screenshot from 2026-05-21 23-44-48.png
Screenshot from 2026-05-22 00-08-48.png
Screenshot from 2026-05-22 00-08-51.png
Screenshot from 2026-05-22 00-11-30.png
Screenshot from 2026-05-22 00-34-00.png
Screenshot from 2026-05-27 19-26-41.png
Screenshot from 2026-05-27 19-26-44.png
Screenshot from 2026-05-27 19-36-18.png
Screenshot from 2026-05-27 19-39-35.png
Screenshot from 2026-05-27 19-50-50.png
Screenshot from 2026-05-27 20-50-14.png
Screenshot from 2026-05-27 22-31-12.png
Screenshot from 2026-06-02 08-11-38.png
Screenshot from 2026-06-03 00-37-40.png
Screenshot from 2026-06-10 06-44-07.png
Screenshot from 2026-06-10 07-09-37.png
xinu
xinu@xinu-develop-end:~$
```

g. Rename file cpuinfo di folder home Anda menjadi infocpu



A terminal window titled 'xinu@xinu-develop-end: ~' with a menu bar (File, Edit, View, Search, Terminal, Help) and a status bar (Wed 10 Jun, 07:20). The terminal shows the execution of two commands: `mv /home/xinu/cpuinfo /home/xinu/infocpu` and `ls /home/xinu/`. The output of the `ls` command lists several files, including README, latihanlinux, prev-version, and a series of screenshot files.

```
xinu
xinu@xinu-develop-end:~$ mv /home/xinu/cpuinfo /home/xinu/infocpu
xinu@xinu-develop-end:~$ ls /home/xinu/
infocpu
latihanlinux
prev-version
README
Screenshot from 2026-03-02 22-16-41.png
Screenshot from 2026-03-02 22-16-43.png
Screenshot from 2026-03-02 22-16-45 - 1.png
Screenshot from 2026-03-02 22-16-45.png
Screenshot from 2026-03-02 22-21-45.png
Screenshot from 2026-03-02 22-24-33.png
Screenshot from 2026-03-02 22-24-36 - 1.png
Screenshot from 2026-03-02 22-24-36.png
Screenshot from 2026-03-02 22-24-37 - 1.png
Screenshot from 2026-03-02 22-24-37 - 2.png
Screenshot from 2026-03-02 22-24-37 - 3.png
Screenshot from 2026-03-02 22-24-37 - 4.png
Screenshot from 2026-03-02 22-24-37.png
Screenshot from 2026-03-02 22-24-38.png
Screenshot from 2026-03-02 22-24-43.png
Screenshot from 2026-03-02 22-24-44.png
Screenshot from 2026-03-02 22-24-46.png
Screenshot from 2026-03-16 15-45-42.png
Screenshot from 2026-03-16 15-45-43.png
Screenshot from 2026-03-16 15-51-01.png
Screenshot from 2026-03-16 15-51-05.png
```



The image shows a terminal window titled "Terminal" with a subtitle "Wed 10 Jun, 07:20". The window contains a list of 25 screenshots, each labeled "Screenshot from" followed by a date and time in YYYY-MM-DD HH-MM-SS format, and a filename ending in ".png". The list starts with "Screenshot from 2026-03-16 16-14-40.png" and ends with "Screenshot from 2026-06-10 07-09-37.png". Below the list, the prompt "xinu" is displayed in blue, followed by "xinu@xinu-develop-end:~\$".

```
Activities Terminal Wed 10 Jun, 07:20 xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
Screenshot from 2026-03-16 16-14-40.png  
Screenshot from 2026-05-21 23-44-44.png  
Screenshot from 2026-05-21 23-44-46 - 1.png  
Screenshot from 2026-05-21 23-44-46.png  
Screenshot from 2026-05-21 23-44-47 - 1.png  
Screenshot from 2026-05-21 23-44-47 - 2.png  
Screenshot from 2026-05-21 23-44-47 - 3.png  
Screenshot from 2026-05-21 23-44-47 - 4.png  
Screenshot from 2026-05-21 23-44-47.png  
Screenshot from 2026-05-21 23-44-48 - 1.png  
Screenshot from 2026-05-21 23-44-48.png  
Screenshot from 2026-05-22 00-08-48.png  
Screenshot from 2026-05-22 00-08-51.png  
Screenshot from 2026-05-22 00-11-30.png  
Screenshot from 2026-05-22 00-34-00.png  
Screenshot from 2026-05-27 19-26-41.png  
Screenshot from 2026-05-27 19-26-44.png  
Screenshot from 2026-05-27 19-36-18.png  
Screenshot from 2026-05-27 19-39-35.png  
Screenshot from 2026-05-27 19-50-50.png  
Screenshot from 2026-05-27 20-50-14.png  
Screenshot from 2026-05-27 22-31-12.png  
Screenshot from 2026-06-02 08-11-38.png  
Screenshot from 2026-06-03 00-37-40.png  
Screenshot from 2026-06-10 06-44-07.png  
Screenshot from 2026-06-10 07-09-37.png  
xinu  
xinu@xinu-develop-end:~$
```

7. Membuat folder baru. (4 poin)

Hint: gunakan mkdir

Buatlah perintah-perintah sehingga dapat melakukan hal-hal berikut ini: (lakukan secara berurutan mulai dengan melakukan perintah cd ~)

- Buatlah folder baru dengan nama "nim_anda".
- Buatlah di dalam folder "nim_anda", folder baru dengan nama "nama_anda".

```
Activities Terminal Wed 10 Jun, 07:24 xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
Screenshot from 2026-06-10 07-09-37.png  
xinu  
xinu@xinu-develop-end:~$ cd ~  
xinu@xinu-develop-end:~$ mkdir 2311104070  
xinu@xinu-develop-end:~$ mkdir 2311104070/AlvinBagusFirmansyah  
xinu@xinu-develop-end:~$ ls  
2311104070  
infocpu  
latihanlinux  
prev-version  
README  
Screenshot from 2026-03-02 22-16-41.png  
Screenshot from 2026-03-02 22-16-43.png  
Screenshot from 2026-03-02 22-16-45 - 1.png  
Screenshot from 2026-03-02 22-16-45.png  
Screenshot from 2026-03-02 22-21-45.png  
Screenshot from 2026-03-02 22-24-33.png  
Screenshot from 2026-03-02 22-24-36 - 1.png  
Screenshot from 2026-03-02 22-24-36.png  
Screenshot from 2026-03-02 22-24-37 - 1.png  
Screenshot from 2026-03-02 22-24-37 - 2.png  
Screenshot from 2026-03-02 22-24-37 - 3.png  
Screenshot from 2026-03-02 22-24-37 - 4.png  
Screenshot from 2026-03-02 22-24-37.png  
Screenshot from 2026-03-02 22-24-38.png  
Screenshot from 2026-03-02 22-24-43.png  
Screenshot from 2026-03-02 22-24-44.png  
Screenshot from 2026-03-02 22-24-46.png
```

The image shows a terminal window titled "Terminal" with the date and time "Wed 10 Jun, 07:24". The user is logged in as "xinu" on a machine named "xinu-develop-end". The terminal displays a list of 25 screenshot files, each starting with "Screenshot from" followed by a date and time in YYYY-MM-DD HH-MM-SS format, and ending with ".png". The files are listed in chronological order from 2026-05-21 23:44:46 to 2026-06-10 07:09:37. Below the list, the user enters the command "ls 2311104070/" and the terminal shows the output "AlvinBagusFirmansyah".

```
Activities Terminal Wed 10 Jun, 07:24 xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
Screenshot from 2026-05-21 23-44-46 - 1.png
Screenshot from 2026-05-21 23-44-46.png
Screenshot from 2026-05-21 23-44-47 - 1.png
Screenshot from 2026-05-21 23-44-47 - 2.png
Screenshot from 2026-05-21 23-44-47 - 3.png
Screenshot from 2026-05-21 23-44-47 - 4.png
Screenshot from 2026-05-21 23-44-47.png
Screenshot from 2026-05-21 23-44-48 - 1.png
Screenshot from 2026-05-21 23-44-48.png
Screenshot from 2026-05-22 00-08-48.png
Screenshot from 2026-05-22 00-08-51.png
Screenshot from 2026-05-22 00-11-30.png
Screenshot from 2026-05-22 00-34-00.png
Screenshot from 2026-05-27 19-26-41.png
Screenshot from 2026-05-27 19-26-44.png
Screenshot from 2026-05-27 19-36-18.png
Screenshot from 2026-05-27 19-39-35.png
Screenshot from 2026-05-27 19-50-50.png
Screenshot from 2026-05-27 20-50-14.png
Screenshot from 2026-05-27 22-31-12.png
Screenshot from 2026-06-02 08-11-38.png
Screenshot from 2026-06-03 00-37-40.png
Screenshot from 2026-06-10 06-44-07.png
Screenshot from 2026-06-10 07-09-37.png
xinu
xinu@xinu-develop-end:~$ ls 2311104070/
AlvinBagusFirmansyah
xinu@xinu-develop-end:~$
```

8. Membaca manual (10 poin)

Hint: gunakan man

- Bukalah fungsi manual untuk perintah "ls"

```
Activities Terminal Wed 10 Jun, 08:00
xinu@xinu-develop-end: /
File Edit View Search Terminal Help
LS(1) User Commands LS(1)
NAME
    ls - list directory contents
SYNOPSIS
    ls [OPTION]... [FILE]...
DESCRIPTION
    List information about the FILES (the current directory by default). Sort
    entries alphabetically if none of -cftuvSUX nor --sort is specified.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        do not ignore entries starting with .

    -A, --almost-all
        do not list implied . and ..

    --author
        with -l, print the author of each file

    -b, --escape
        print C-style escapes for nongraphic characters

    --block-size=SIZE
Manual page ls(1) line 1 (press h for help or q to quit)
```



```
Activities Terminal Wed 10 Jun, 08:00
xinu@xinu-develop-end: /
File Edit View Search Terminal Help
-B, --ignore-backups
    do not list implied entries ending with ~

-c      with -lt: sort by, and show, ctime (time of last modification of file
        status information); with -l: show ctime and sort by name; otherwise:
        sort by ctime, newest first

-C      list entries by columns

--color[=WHEN]
        colorize the output; WHEN can be 'never', 'auto', or 'always' (the
        default); more info below

-d, --directory
        list directories themselves, not their contents

-D, --dired
        generate output designed for Emacs' dired mode

-f      do not sort, enable -aU, disable -ls --color

-F, --classify
        append indicator (one of */=>@|) to entries

--file-type
        likewise, except do not append '*'

Manual page ls(1) line 33 (press h for help or q to quit)
```

```
Activities Terminal Wed 10 Jun, 08:01
xinu@xinu-develop-end: /
File Edit View Search Terminal Help

--file-type
    likewise, except do not append '*'

--format=WORD
    across -x, commas -m, horizontal -x, long -l, single-column -l, ver-
    bose -l, vertical -C

--full-time
    like -l --time-style=full-iso

-g    like -l, but do not list owner

--group-directories-first
    group directories before files;

    can be augmented with a --sort option, but any use of --sort=none (-U)
    disables grouping

-G, --no-group
    in a long listing, don't print group names

-h, --human-readable
    with -l and/or -s, print human readable sizes (e.g., 1K 234M 2G)

--si   likewise, but use powers of 1000 not 1024

-H, --dereference-command-line

Manual page ls(1) line 57 (press h for help or q to quit)
```

```
Activities Terminal Wed 10 Jun, 08:01 xinu@xinu-develop-end: /
```

File Edit View Search Terminal Help

```
--si likewise, but use powers of 1000 not 1024

-H, --dereference-command-line
    follow symbolic links listed on the command line

--dereference-command-line-symlink-to-dir
    follow each command line symbolic link
    that points to a directory

--hide=PATTERN
    do not list implied entries matching shell PATTERN (overridden by -a
    or -A)

--indicator-style=WORD
    append indicator with style WORD to entry names: none (default), slash
    (-p), file-type (--file-type), classify (-F)

-i, --inode
    print the index number of each file

-I, --ignore=PATTERN
    do not list implied entries matching shell PATTERN

-k, --kibibytes
    default to 1024-byte blocks for disk usage
```

Manual page ls(1) line 81 (press h for help or q to quit)

```
Activities [Terminal] Wed 10 Jun, 08:01
xinu@xinu-develop-end: /
File Edit View Search Terminal Help
  -l      use a long listing format

  -L, --dereference
           when showing file information for a symbolic link, show information
           for the file the link references rather than for the link itself

  -m      fill width with a comma separated list of entries

  -n, --numeric-uid-gid
           like -l, but list numeric user and group IDs

  -N, --literal
           print raw entry names (don't treat e.g. control characters specially)

  -o      like -l, but do not list group information

  -p, --indicator-style=slash
           append / indicator to directories

  -q, --hide-control-chars
           print ? instead of nongraphic characters

  --show-control-chars
           show nongraphic characters as-is (the default, unless program is 'ls'
           and output is a terminal)

  -Q, --quote-name
Manual page ls(1) line 108 (press h for help or q to quit)
```

```
Activities [Terminal] Wed 10 Jun, 08:01
xinu@xinu-develop-end: /
File Edit View Search Terminal Help

-r, --reverse
    reverse order while sorting

-R, --recursive
    list subdirectories recursively

-s, --size
    print the allocated size of each file, in blocks

-S
    sort by file size

--sort=WORD
    sort by WORD instead of name: none (-U), size (-S), time (-t), version
    (-v), extension (-X)

--time=WORD
    with -l, show time as WORD instead of default modification time: atime
    or access or use (-u) ctime or status (-c); also use specified time as
    sort key if --sort=time

--time-style=STYLE
    with -l, show times using style STYLE: full-iso, long-iso, iso,
    locale, or +FORMAT; FORMAT is interpreted like in 'date'; if FORMAT is
    FORMAT1<newline>FORMAT2, then FORMAT1 applies to non-recent files and
    FORMAT2 to recent files; if STYLE is prefixed with 'posix-', STYLE
    takes effect only outside the POSIX locale

Manual page ls(1) line 141 (press h for help or q to quit)
```

```
Activities [Terminal] Wed 10 Jun, 08:01
xinu@xinu-develop-end: /
File Edit View Search Terminal Help
  -t      sort by modification time, newest first
  -T, --tabsize=COLS
           assume tab stops at each COLS instead of 8
  -u      with -lt: sort by, and show, access time; with -l: show access time
           and sort by name; otherwise: sort by access time
  -U      do not sort; list entries in directory order
  -v      natural sort of (version) numbers within text
  -w, --width=COLS
           assume screen width instead of current value
  -x      list entries by lines instead of by columns
  -X      sort alphabetically by entry extension
  -Z, --context
           print any security context of each file
  -l      list one file per line
  --help  display this help and exit
  --version
Manual page ls(1) line 168 (press h for help or q to quit)
```

```
Activities Terminal Wed 10 Jun, 08:02 xinu@xinu-develop-end: /
File Edit View Search Terminal Help
Units are K,M,G,T,P,E,Z,Y (powers of 1024) or KB,MB,... (powers of 1000).

Using color to distinguish file types is disabled both by default and with
--color=never. With --color=auto, ls emits color codes only when standard
output is connected to a terminal. The LS_COLORS environment variable can
change the settings. Use the dircolors command to set it.

Exit status:
0      if OK,

1      if minor problems (e.g., cannot access subdirectory),

2      if serious trouble (e.g., cannot access command-line argument).

AUTHOR
Written by Richard M. Stallman and David MacKenzie.

REPORTING BUGS
GNU coreutils online help: <http://www.gnu.org/software/coreutils/>
Report ls translation bugs to <http://translationproject.org/team/>

COPYRIGHT
Copyright © 2014 Free Software Foundation, Inc. License GPLv3+: GNU GPL ver-
sion 3 or later <http://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it. There is
NO WARRANTY, to the extent permitted by law.

Manual page ls(1) line 198 (press h for help or q to quit)
```

```
Activities Terminal Wed 10 Jun, 08:02
xinu@xinu-develop-end: /

File Edit View Search Terminal Help
change the settings. Use the dircolors command to set it.

Exit status:
0      if OK,
1      if minor problems (e.g., cannot access subdirectory),
2      if serious trouble (e.g., cannot access command-line argument).

AUTHOR
Written by Richard M. Stallman and David MacKenzie.

REPORTING BUGS
GNU coreutils online help: <http://www.gnu.org/software/coreutils/>
Report ls translation bugs to <http://translationproject.org/team/>

COPYRIGHT
Copyright © 2014 Free Software Foundation, Inc. License GPLv3+: GNU GPL ver-
sion 3 or later <http://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it. There is
NO WARRANTY, to the extent permitted by law.

SEE ALSO
Full documentation at: <http://www.gnu.org/software/coreutils/ls>
or available locally via: info '(coreutils) ls invocation'

GNU coreutils 8.23 March 2015 LS(1)
Manual page ls(1) line 203/231 (END) (press h for help or q to quit)
```

- b. Apa fungsi perintah “ls”?
Perintah ls digunakan untuk menampilkan daftar file dan direktori pada suatu folder.
- c. Siapakah pencipta perintah “ls”?
Richard M. Stallman dan David MacKenzie.
- d. Apakah arti dari -h dari manual ls?
Menampilkan ukuran file dalam format yang mudah dibaca manusia (KB, MB, GB).
- e. Option apa yang harus digunakan agar dapat melihat direktori secara rekursif?
Option:
-R
Contoh:
ls -R
Fungsinya menampilkan isi direktori beserta seluruh subdirektori secara rekursif.
- f. Bukalah fungsi manual untuk perintah “cp”


```
Activities Terminal Wed 10 Jun, 08:05 xinu@xinu-develop-end: /
File Edit View Search Terminal Help
CP(1) User Commands CP(1)

NAME
    cp - copy files and directories

SYNOPSIS
    cp [OPTION]... [-I] SOURCE DEST
    cp [OPTION]... SOURCE... DIRECTORY
    cp [OPTION]... -t DIRECTORY SOURCE...

DESCRIPTION
    Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --archive
        same as -dR --preserve=all

    --attributes-only
        don't copy the file data, just the attributes

    --backup[=CONTROL]
        make a backup of each existing destination file

    -b
        like --backup but does not accept an argument

    --copy-content
Manual page cp(1) line 1 (press h for help or q to quit)
```

```
Activities > Terminal Wed 10 Jun, 08:05
xinu@xinu-develop-end: /
File Edit View Search Terminal Help
copy contents of special files when recursive

-d same as --no-dereference --preserve=links

-f, --force
    if an existing destination file cannot be opened, remove it and try
    again (this option is ignored when the -n option is also used)

-i, --interactive
    prompt before overwrite (overrides a previous -n option)

-H follow command-line symbolic links in SOURCE

-l, --link
    hard link files instead of copying

-L, --dereference
    always follow symbolic links in SOURCE

-n, --no-clobber
    do not overwrite an existing file (overrides a previous -i option)

-P, --no-dereference
    never follow symbolic links in SOURCE

-p same as --preserve=mode,ownership,timestamps

Manual page cp(1) line 30 (press h for help or q to quit)
```

```
Activities [Terminal] Wed 10 Jun, 08:06
xinu@xinu-develop-end: /
File Edit View Search Terminal Help

--preserve[=ATTR_LIST]
    preserve the specified attributes (default: mode,ownership,time-
    stamps), if possible additional attributes: context, links, xattr, all

--no-preserve=ATTR_LIST
    don't preserve the specified attributes

--parents
    use full source file name under DIRECTORY

-R, -r, --recursive
    copy directories recursively

--reflink[=WHEN]
    control clone/CoW copies. See below

--remove-destination
    remove each existing destination file before attempting to open it
    (contrast with --force)

--sparse=WHEN
    control creation of sparse files. See below

--strip-trailing-slashes
    remove any trailing slashes from each SOURCE argument

-s, --symbolic-link
Manual page cp(1) line 57 (press h for help or q to quit)
```

File Edit View Search Terminal Help

override the usual backup suffix

-t, --target-directory=DIRECTORY

copy all SOURCE arguments into DIRECTORY

-T, --no-target-directory

treat DEST as a normal file

-u, --update

copy only when the SOURCE file is newer than the destination file or when the destination file is missing

-v, --verbose

explain what is being done

-x, --one-file-system

stay on this file system

-Z

set SELinux security context of destination file to default type

--context[=CTX**]**like **-Z**, or if CTX is specified then set the SELinux or SMACK security context to CTX**--help** display this help and exit**--version**

Manual page cp(1) line 87 (press h for help or q to quit)

```
Activities Terminal Wed 10 Jun, 08:06 xinu@xinu-develop-end: /
```

File Edit View Search Terminal Help

output version information and exit

By default, sparse SOURCE files are detected by a crude heuristic and the corresponding DEST file is made sparse as well. That is the behavior selected by **--sparse=auto**. Specify **--sparse=always** to create a sparse DEST file whenever the SOURCE file contains a long enough sequence of zero bytes. Use **--sparse=never** to inhibit creation of sparse files.

When **--reflink[=always]** is specified, perform a lightweight copy, where the data blocks are copied only when modified. If this is not possible the copy fails, or if **--reflink=auto** is specified, fall back to a standard copy.

The backup suffix is '~', unless set with **--suffix** or SIMPLE_BACKUP_SUFFIX. The version control method may be selected via the **--backup** option or through the VERSION_CONTROL environment variable. Here are the values:

- none, off
never make backups (even if **--backup** is given)
- numbered, t
make numbered backups
- existing, nil
numbered if numbered backups exist, simple otherwise
- simple, never
always make simple backups

Manual page cp(1) line 114 (press h for help or q to quit)

```
Activities Terminal Wed 10 Jun, 08:06 xinu@xinu-develop-end: /
File Edit View Search Terminal Help
again (this option is ignored when the -n option is also used)

-i, --interactive
    prompt before overwrite (overrides a previous -n option)

-H      follow command-line symbolic links in SOURCE

-l, --link
    hard link files instead of copying

-L, --dereference
    always follow symbolic links in SOURCE

-n, --no-clobber
    do not overwrite an existing file (overrides a previous -i option)

-P, --no-dereference
    never follow symbolic links in SOURCE

-p      same as --preserve=mode,ownership,timestamps

--preserve[=ATTR_LIST]
    preserve the specified attributes (default: mode,ownership,time-
    stamps), if possible additional attributes: context, links, xattr, all

--no-preserve=ATTR_LIST
    don't preserve the specified attributes
Manual page cp(1) line 36 (press h for help or q to quit)
```

```
Activities Terminal Wed 10 Jun, 08:07 xinu@xinu-develop-end: /
File Edit View Search Terminal Help

--remove-destination
    remove each existing destination file before attempting to open it
    (contrast with --force)

--sparse=WHEN
    control creation of sparse files. See below

--strip-trailing-slashes
    remove any trailing slashes from each SOURCE argument

-s, --symbolic-link
    make symbolic links instead of copying

-S, --suffix=SUFFIX
    override the usual backup suffix

-t, --target-directory=DIRECTORY
    copy all SOURCE arguments into DIRECTORY

-T, --no-target-directory
    treat DEST as a normal file

-u, --update
    copy only when the SOURCE file is newer than the destination file or
    when the destination file is missing

Manual page cp(1) line 72 (press h for help or q to quit)
```

```
Activities [Terminal] Wed 10 Jun, 08:07
xinu@xinu-develop-end: /
File Edit View Search Terminal Help

-v, --verbose
    explain what is being done

-x, --one-file-system
    stay on this file system

-Z      set SELinux security context of destination file to default type

--context[=CTX]
    like -Z, or if CTX is specified then set the SELinux or SMACK security
    context to CTX

--help display this help and exit

--version
    output version information and exit

By default, sparse SOURCE files are detected by a crude heuristic and the
corresponding DEST file is made sparse as well. That is the behavior
selected by --sparse=auto. Specify --sparse=always to create a sparse DEST
file whenever the SOURCE file contains a long enough sequence of zero bytes.
Use --sparse=never to inhibit creation of sparse files.

When --reflink=[always] is specified, perform a lightweight copy, where the
data blocks are copied only when modified. If this is not possible the copy
fails, or if --reflink=auto is specified, fall back to a standard copy.

Manual page cp(1) line 99 (press h for help or q to quit)
```



```
Activities Terminal Wed 10 Jun, 08:07
xinu@xinu-develop-end: /

File Edit View Search Terminal Help

    numbered if numbered backups exist, simple otherwise

    simple, never
    always make simple backups

    As a special case, cp makes a backup of SOURCE when the force and backup
    options are given and SOURCE and DEST are the same name for an existing, reg-
    ular file.

AUTHOR
    Written by Torbjorn Granlund, David MacKenzie, and Jim Meyering.

REPORTING BUGS
    GNU coreutils online help: <http://www.gnu.org/software/coreutils/>
    Report cp translation bugs to <http://translationproject.org/team/>

COPYRIGHT
    Copyright © 2014 Free Software Foundation, Inc. License GPLv3+: GNU GPL ver-
    sion 3 or later <http://gnu.org/licenses/gpl.html>.
    This is free software: you are free to change and redistribute it. There is
    NO WARRANTY, to the extent permitted by law.

SEE ALSO
    Full documentation at: <http://www.gnu.org/software/coreutils/cp>
    or available locally via: info '(coreutils) cp invocation'

GNU coreutils 8.23 March 2015 CP(1)
Manual page cp(1) line 137/165 (END) (press h for help or q to quit)
```

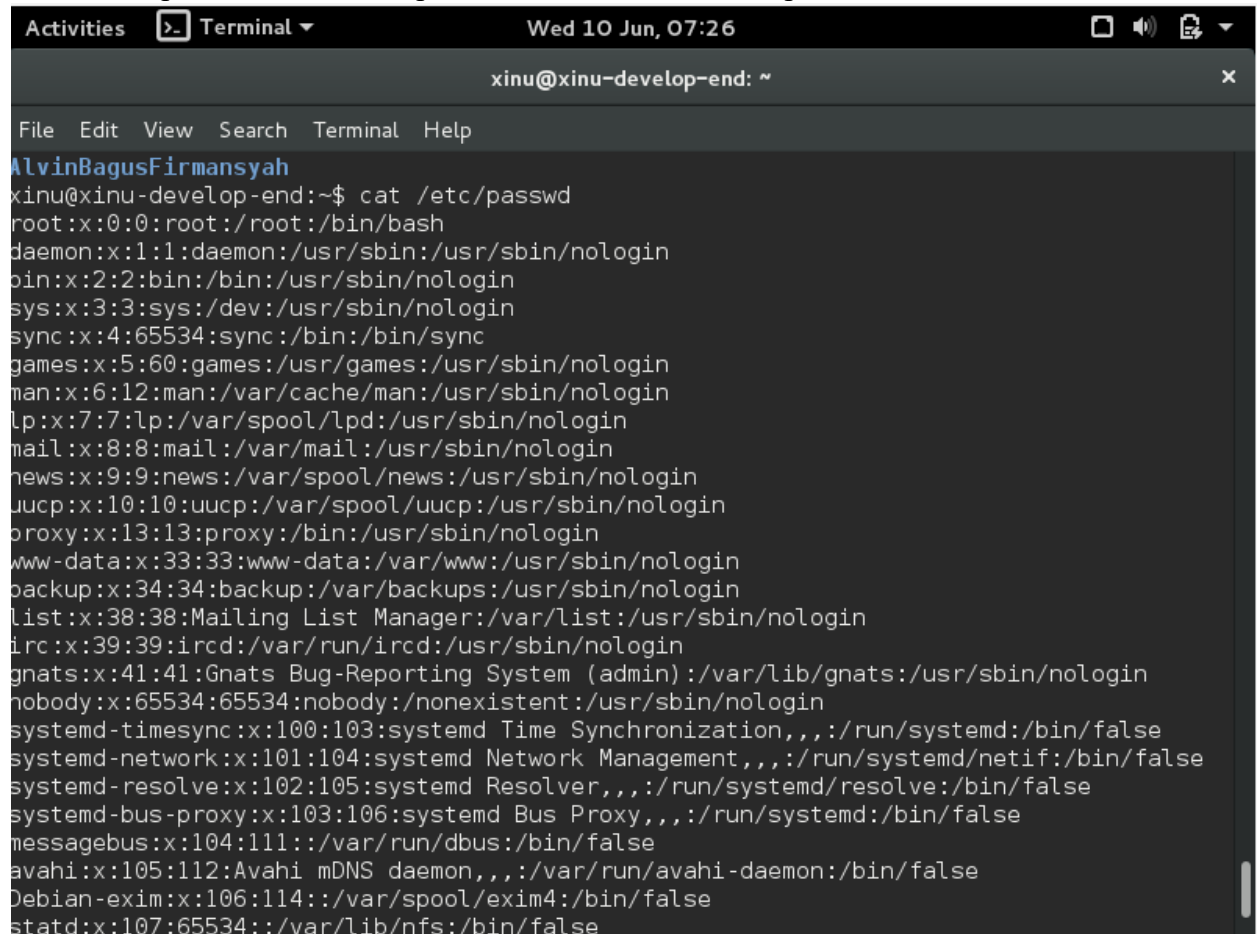
- g. Apa fungsi perintah “cp”
Perintah cp digunakan untuk menyalin (copy) file atau direktori.
- h. Siapakah pencipta perintah “cp”?

Torbjorn Granlund, David MacKenzie, dan Jim Meyering.

- i. Apakah arti -v dalam perintah “cp”
Fungsinya menampilkan proses penyalinan secara detail.
- j. Jika ingin interaktif, option apa yang harus digunakan?
Fungsinya meminta konfirmasi sebelum menerima file yang sudah ada.

9. Pipe (12 poin)

a. Lakukan perintah ini `cat /etc/passwd` dan screenshot hasil perintah tersebut!



```
Activities Terminal Wed 10 Jun, 07:26
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
AlvinBagusFirmansyah
xinu@xinu-develop-end:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:100:103:systemd Time Synchronization,,,:/run/systemd:/bin/false
systemd-network:x:101:104:systemd Network Management,,,:/run/systemd/netif:/bin/false
systemd-resolve:x:102:105:systemd Resolver,,,:/run/systemd/resolve:/bin/false
systemd-bus-proxy:x:103:106:systemd Bus Proxy,,,:/run/systemd:/bin/false
messagebus:x:104:111:/:/var/run/dbus:/bin/false
avahi:x:105:112:Avahi mDNS daemon,,,:/var/run/avahi-daemon:/bin/false
Debian-exim:x:106:114:/:/var/spool/exim4:/bin/false
statd:x:107:65534:/:/var/lib/nfs:/bin/false
```

b. Apa fungsi perintah `cat`?

Perintah `cat` (concatenate) digunakan untuk:

- Menampilkan isi file ke terminal.
- Menggabungkan beberapa file.
- Membuat file sederhana.

Pada perintah:

`cat /etc/passwd`

digunakan untuk menampilkan isi file `/etc/passwd` yang berisi informasi akun pengguna pada sistem Linux.

c. Lakukan perintah `cat /etc/passwd | grep daemon` dan screenshot hasil perintah tersebut!

```
xinu@xinu-develop-end:~$ cat /etc/passwd | grep daemon
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
avahi:x:105:112:Avahi mDNS daemon,,,:/var/run/avahi-daemon:/bin/false
avahi-autoipd:x:108:117:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/bin/false
colord:x:109:119:colord colour management daemon,,,:/var/lib/colord:/bin/false
pulse:x:112:122:PulseAudio daemon,,,:/var/run/pulse:/bin/false
usbmux:x:116:46:usbmux daemon,,,:/var/lib/usbmux:/bin/false
tftp:x:120:127:tftp daemon,,,:/srv/tftp:/bin/false
xinu@xinu-develop-end:~$
```

d. Lakukan perintah `cat /etc/passwd | grep root` dan screenshot hasil perintah tersebut!

```
xinu@xinu-develop-end:~$ cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash
xinu@xinu-develop-end:~$
```

e. Lakukan perintah `cat /etc/passwd | grep nobody` dan screenshot hasil perintah tersebut!

```
xinu@xinu-develop-end:~$ cat /etc/passwd | grep nobody
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
xinu@xinu-develop-end:~$
```

f. Apakah fungsi perintah “`| grep daemon`”?

Tanda pipe (`|`) mengirimkan output dari perintah sebelumnya ke perintah berikutnya.

Pada:

`cat /etc/passwd | grep daemon`

- `cat /etc/passwd` menampilkan seluruh isi file.
- `grep daemon` mencari baris yang mengandung kata `daemon`.

Jadi perintah tersebut berfungsi untuk menampilkan hanya baris yang mengandung kata `daemon`.

10. Redirection (14 poin)

a. Lakukan perintah dan jelaskan hasilnya

`cd /`

`ls -al > /home/user/result.txt`

Ganti user dengan username ubuntu anda.

```
xinu@xinu-develop-end:/$ cd /
xinu@xinu-develop-end:/$ ls -al > /home/xinu/result.txt
xinu@xinu-develop-end:/$ /home/xinu/result.txt
bash: /home/xinu/result.txt: Permission denied
xinu@xinu-develop-end:/$
```

```
xinu@xinu-develop-end:/$ ls -al > /home/xinu/result.txt
xinu@xinu-develop-end:/$ ls /home/xinu
2311104070
infocpu
latihanlinux
prev-version
README
result.txt
Screenshot from 2026-03-02 22-16-41.png
Screenshot from 2026-03-02 22-16-43.png
Screenshot from 2026-03-02 22-16-45 - 1.png
Screenshot from 2026-03-02 22-16-45.png
Screenshot from 2026-03-02 22-21-45.png
Screenshot from 2026-03-02 22-24-33.png
Screenshot from 2026-03-02 22-24-36 - 1.png
Screenshot from 2026-03-02 22-24-36.png
Screenshot from 2026-03-02 22-24-37 - 1.png
Screenshot from 2026-03-02 22-24-37 - 2.png
Screenshot from 2026-03-02 22-24-37 - 3.png
Screenshot from 2026-03-02 22-24-37 - 4.png
Screenshot from 2026-03-02 22-24-37.png
Screenshot from 2026-03-02 22-24-38.png
Screenshot from 2026-03-02 22-24-43.png
```

b. Dimana file result.txt berada?

```
xinu@xinu-develop-end:/$ ls -al > /home/xinu/result.txt
xinu@xinu-develop-end:/$ ls /home/xinu
2311104070
infocpu
latihanlinux
prev-version
README
result.txt
Screenshot from 2026-03-02 22-16-41.png
Screenshot from 2026-03-02 22-16-43.png
Screenshot from 2026-03-02 22-16-45 - 1.png
Screenshot from 2026-03-02 22-16-45.png
Screenshot from 2026-03-02 22-21-45.png
Screenshot from 2026-03-02 22-24-33.png
Screenshot from 2026-03-02 22-24-36 - 1.png
Screenshot from 2026-03-02 22-24-36.png
Screenshot from 2026-03-02 22-24-37 - 1.png
Screenshot from 2026-03-02 22-24-37 - 2.png
Screenshot from 2026-03-02 22-24-37 - 3.png
Screenshot from 2026-03-02 22-24-37 - 4.png
Screenshot from 2026-03-02 22-24-37.png
Screenshot from 2026-03-02 22-24-38.png
Screenshot from 2026-03-02 22-24-43.png
```

c. Lakukan perintah dan jelaskan hasilnya

cd /etc

ls -al > /home/user/result.txt

Ganti user dengan username ubuntu anda.

```
xinu@xinu-develop-end:/$ cd /etc
xinu@xinu-develop-end:/etc$ ls -al > /home/xinu/result.txt
xinu@xinu-develop-end:/etc$
```

d. Apakah fungsi dari perintah >?

Operator:

>

berfungsi untuk:

- Mengalihkan output ke file.
- Membuat file baru jika belum ada.
- Menimpa isi file lama jika file sudah ada.

e. Lakukan perintah dan jelaskan hasilnya

```
xinu@xinu-develop-end:/etc$ cd /
xinu@xinu-develop-end:/$ ls -al >> /home/xinu/result.txt
xinu@xinu-develop-end:/$
```

Output direktori root (/) disimpan ke file result1.txt.

Jika file belum ada maka akan dibuat.

11. Kompile Source Code (Total 17 Poin)

Pastikan gcc sudah terinstall pada Ubuntu anda menggunakan gcc --version

Panggil atau hubungi asisten praktikum apabila gcc belum terinstall!

Hint:

gcc -o output file

./

Screenshot setiap tahapan yang anda lakukan terhadap Ubuntu!

1. Buatlah file dengan nama 2_1.c yang berisi:

```
2. #include <stdio.h> //librari untuk printf()
3. int main(){ //main()
4. int a = 1; // variabel a diinisiasi dengan 1 <stdio
5. int b = 2; // variabel b diinisiasi dengan 2
6. printf("Hello World!\n"); // tampilkan "Hello World" pada
7. layar
8. printf("a = %d, b = %d, a + b = %d\n", a, b, a+b);
9. // tampilkan a = 1, b = 2, a + b = 3
10. }
```

2. Kompile source code tersebut menggunakan gcc! Nama output program adalah

2_1

(bukan a.out). Tuliskan perintah untuk mengkompile source code tersebut!

3. Jalankan program yang baru saja Anda compile. Tuliskan perintah untuk menjalankan

program tersebut!

4. Buatlah file dengan nama 2_2.c yang berisi:

```
#include <stdio.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <fcntl.h>
#include <string.h>
#include <errno.h>
int main(int argc, char *argv[])
{
    int fd;
    if(2 != argc){
        printf("\n Penggunaan 'myopen nama_file' \n");
        return 1;
    }
    errno = 0;
    fd = open(argv[1],O_RDONLY);
    if(-1 == fd){
        printf("\n myopen gagal membuka file %s dengan pesan
error:
[%s]\n", argv[1],strerror(errno));
        return 1;
    }else{
        printf("\n myopen berhasil membuka file %s\n",
argv[1]);
    }
    return 0;
}
```

5. Kompilasi source code tersebut menggunakan gcc! Nama output program adalah “myopen”. Tulis perintah untuk mengkompilasi source code tersebut.

6. Jalankan program myopen yang baru saja Anda buat! Tuliskan perintah untuk menjalankan program myopen.

7. Jelaskan apa yang dilakukan program tersebut!

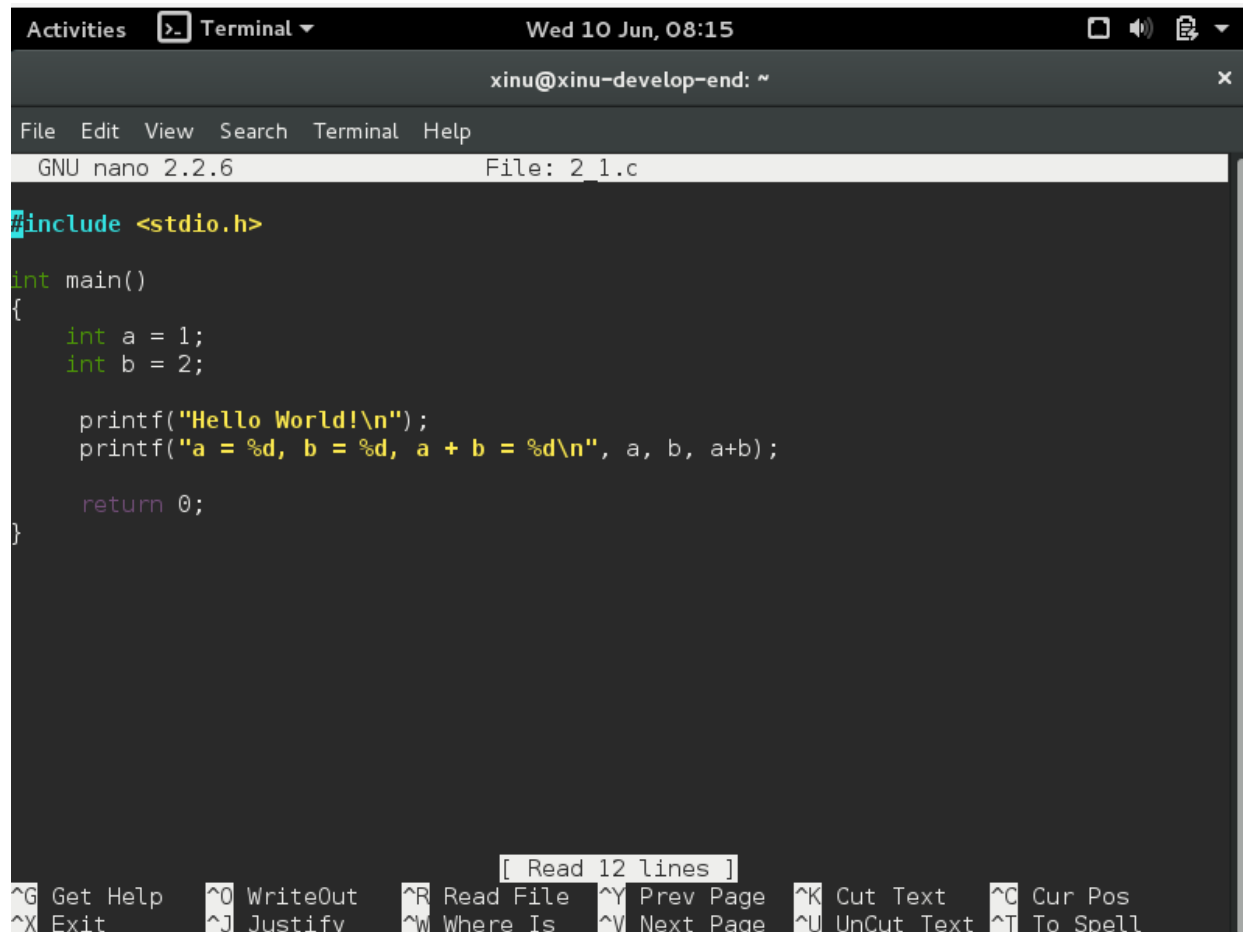
Jawab:

1. Cek gcc sudah terinstall

```
xinu@xinu-develop-end:/$ gcc --version
gcc (Debian 4.9.2-10) 4.9.2
Copyright (C) 2014 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

xinu@xinu-develop-end:/$
```

2. Buat file 2_1.c



```
Activities Terminal Wed 10 Jun, 08:15
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
GNU nano 2.2.6 File: 2_1.c

#include <stdio.h>

int main()
{
    int a = 1;
    int b = 2;

    printf("Hello World!\n");
    printf("a = %d, b = %d, a + b = %d\n", a, b, a+b);

    return 0;
}

[ Read 12 lines ]
^G Get Help  ^O WriteOut  ^R Read File ^Y Prev Page ^K Cut Text   ^C Cur Pos
^X Exit      ^J Justify   ^W Where Is  ^V Next Page ^U UnCut Text ^T To Spell
```

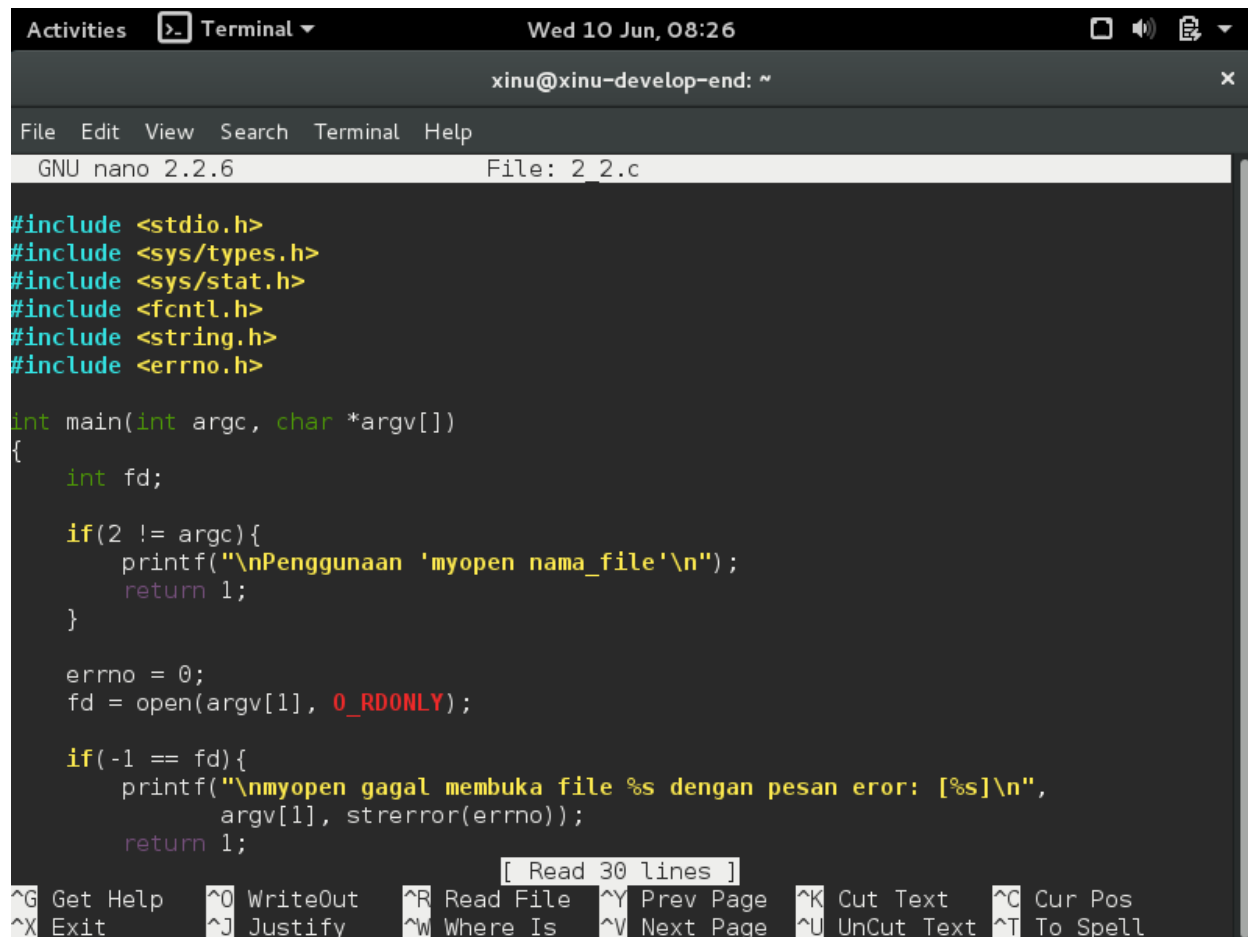
3. Kompilasi 2_1.c

```
xinu@xinu-develop-end:~$ gcc -o 2_1 2_1.c
xinu@xinu-develop-end:~$ gcc -o 2_1 2_1.c
xinu@xinu-develop-end:~$ ./2_1
Hello World!
a = 1, b = 2, a + b = 3
xinu@xinu-develop-end:~$
```

4. Jalankan program

```
xinu@xinu-develop-end:~$ gcc -o 2_1 2_1.c
xinu@xinu-develop-end:~$ gcc -o 2_1 2_1.c
xinu@xinu-develop-end:~$ ./2_1
Hello World!
a = 1, b = 2, a + b = 3
xinu@xinu-develop-end:~$
```

5. Buat file 2_2.c



```
Activities Terminal Wed 10 Jun, 08:26
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
GNU nano 2.2.6 File: 2_2.c

#include <stdio.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <fcntl.h>
#include <string.h>
#include <errno.h>

int main(int argc, char *argv[])
{
    int fd;

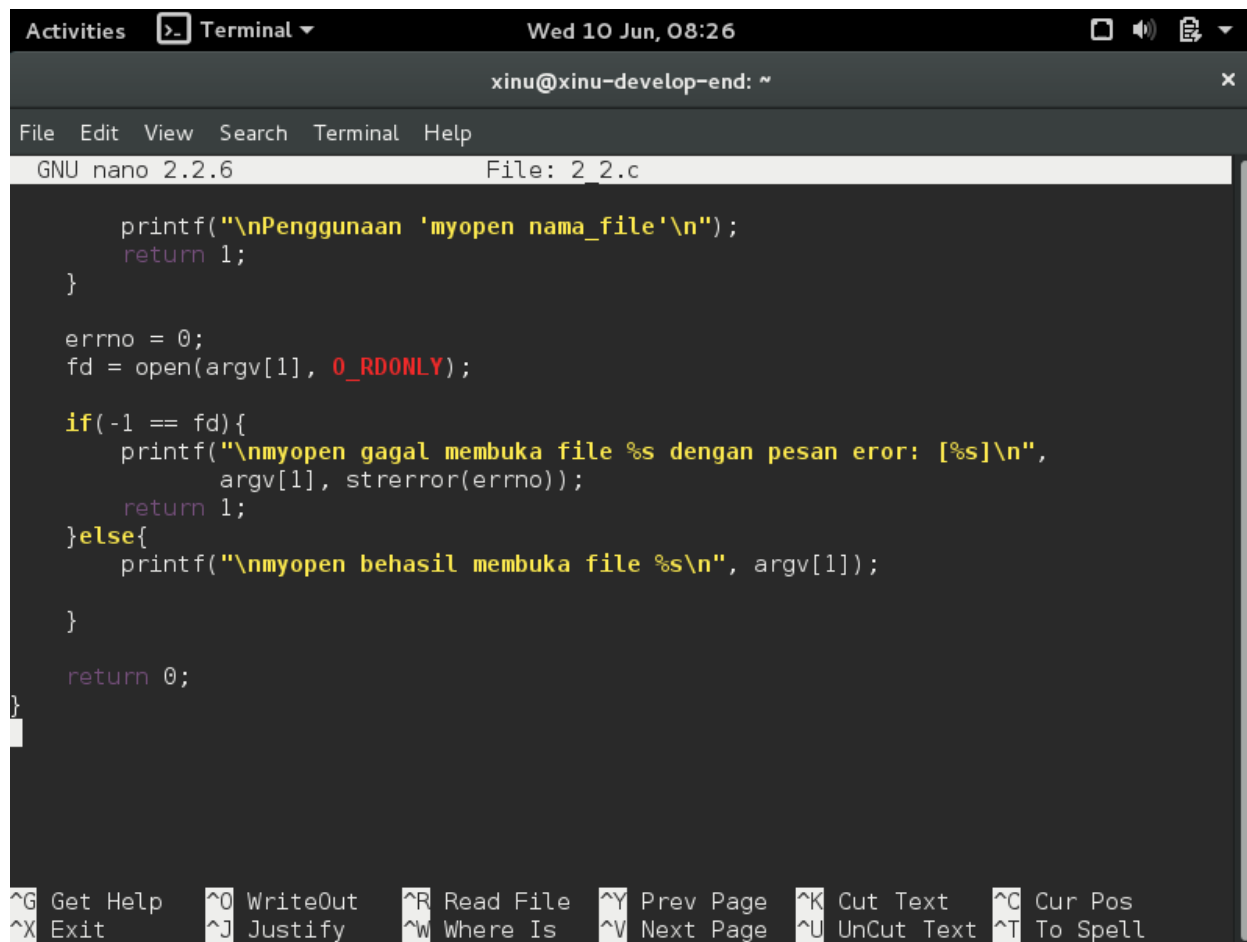
    if(2 != argc){
        printf("\nPenggunaan 'myopen nama_file'\n");
        return 1;
    }

    errno = 0;
    fd = open(argv[1], O_RDONLY);

    if(-1 == fd){
        printf("\nmyopen gagal membuka file %s dengan pesan error: [%s]\n",
            argv[1], strerror(errno));
        return 1;
    }
}
```

[Read 30 lines]

^G Get Help	^O WriteOut	^R Read File	^Y Prev Page	^K Cut Text	^C Cur Pos
^X Exit	^J Justify	^W Where Is	^V Next Page	^U UnCut Text	^T To Spell



```
GNU nano 2.2.6 File: 2_2.c

    printf("\nPenggunaan 'myopen nama_file'\n");
    return 1;
}

errno = 0;
fd = open(argv[1], O_RDONLY);

if(-1 == fd){
    printf("\nmyopen gagal membuka file %s dengan pesan error: [%s]\n",
        argv[1], strerror(errno));
    return 1;
}else{
    printf("\nmyopen berhasil membuka file %s\n", argv[1]);
}

return 0;
}
```

6. Kompilasi menjadi myopen

```
xinu@xinu-develop-end:~$ gcc -o myopen 2_2.c
xinu@xinu-develop-end:~$
```

7. Jalankan program myopen

```
xinu@xinu-develop-end:~$ ./myopen 2_1.c

myopen berhasil membuka file 2_1.c
xinu@xinu-develop-end:~$
```

8. Jawaban nomor 7 (Penjelasan Program)

Program myopen digunakan untuk membuka file yang diberikan sebagai argumen pada command line. Program akan memeriksa apakah file tersebut dapat dibuka dengan mode read-only (O_RDONLY).

- Jika file berhasil dibuka, program menampilkan pesan bahwa file berhasil dibuka.
- Jika file gagal dibuka, program menampilkan pesan error yang sesuai menggunakan strerror(errno).

Dengan kata lain, program ini berfungsi untuk mengecek keberadaan dan akses suatu file serta menampilkan status berhasil atau gagal saat membuka file tersebut.

BAB III

PENUTUP

C. Kesimpulan

Pada praktikum ini saya mempelajari berbagai perintah dasar Linux seperti navigasi direktori, pengelolaan file dan folder, penggunaan manual command, pipe, dan redirection. Selain itu, saya juga mempelajari cara membuat, menyalin, menghapus, serta mengubah nama file melalui terminal.

